Question 1 (OSI Model)

Which layer of the OSI model is responsible for the physical transmission of raw data bits over a communication medium?

Answer Choices:

A. Transport Layer

B. Data Link Layer

C. Physical Layer

D. Network Layer

Question 2 (TCP/IP Model)

Which layer in the TCP/IP model combines the responsibilities of the OSI model's Presentation and Session layers with the Application layer?

Answer Choices:

A. Internet Layer

B. Transport Layer

C. Network Access Layer

D. Application Layer

Question 3 (Ethernet)

What is the primary function of the Carrier Sense Multiple Access with Collision Detection (CSMA/CD) mechanism used in traditional Ethernet networks?

Answer Choices:

A. To guarantee reliable, connection-oriented data delivery.

B. To manage the physical addresses (MAC addresses) on the network.

C. To listen for traffic and detect simultaneous transmissions on a shared medium.

D. To provide logical addressing (IP addresses).

Question 4 (Networking Cables)

The "e" in **CAT5e** (Category 5 enhanced) UTP cabling primarily indicates an enhancement in performance to reduce:

Answer Choices:

A. Physical size

B. Crosstalk

C. Cost

D. Fire hazards

Question 5 (TCP vs. UDP)

Which of the following protocols is classified as **connectionless** and does not guarantee delivery or sequencing of packets?

Answer Choices:

A. FTP

B. TCP

C. HTTP

D. UDP

Question 6 (IP Subnetting)

For the Class C IP address $192.168.1.0$ using the default subnet mask, how many bits are used for the Network part?

Answer Choices:

A. 8 bits

B. 16 bits

C. 24 bits

D. 32 bits

Question 7 (Router IOS)

When managing a Cisco router, what type of software interface allows a user to directly interact with the device using command lines?

Answer Choices:

A. SDM (Security Device Manager)

B. ROM Monitor (ROMMON)

C. IOS (Internetworking Operating System) CLI

D. NVRAM

Question 8 (VLANs)

The main purpose of a **Virtual LAN (VLAN)** is to segment a single physical switch into multiple logical:

Answer Choices:

A. Collision Domains

B. Routing Tables

C. Broadcast Domains

D. Physical Connections

Question 9 (Storage Basics)

Which storage type refers to storage devices that are physically attached directly to a single server or host and are not shared over a network?

Answer Choices:

A. NAS (Network Attached Storage)

B. SAN (Storage Area Network)

C. Parallel File System

D. DAS (Direct Attached Storage)

Question 10 (Parallel File System)

Which of the following is an example of a **Parallel File System** designed for high-performance computing (HPC) clusters?

Answer Choices:

A. FAT32

B. Ceph

C. NTFS

D. Ext4

Intermediate-Level Multiple-Choice Questions (20 Questions)

Question 11 (OSI Model)

Which layer of the OSI model is responsible for logically addressing devices (IP addressing) and determining the best path for data across interconnected networks?

Answer Choices:

A. Transport Layer

B. Network Layer

C. Data Link Layer

D. Session Layer

Question 12 (UTP vs. STP)

What is the primary difference in construction between **Unshielded Twisted Pair (UTP)** and **Shielded Twisted Pair (STP)** cable?

Answer Choices:

A. STP uses thicker copper wires.

B. UTP has a foil or braided jacket surrounding the wire pairs.

C. STP includes a metallic foil or braid layer for EMI/RFI noise rejection.

D. UTP supports higher transmission speeds.

Question 13 (IP Subnetting)

A network administrator uses the IP address $192.168.5.85$ with a subnet mask of $/24$. What is the **network** portion of this address?

Answer Choices:

A. $192.168.5.0$

B. $192.168.0.0$

C. $192.168.5.85$

D. $0.0.0.85$

Question 14 (Subnetting)

Which subnet mask is required to divide a network and support exactly 52 usable hosts in a subnet?

Answer Choices:

A. $255.255 .255.192$

B. $255.255 .255.224$

C. $255.255 .255.240$

D. $255.255 .255.252$

Question 15 (Routing Protocols)

What is the primary characteristic that defines a **Distance Vector** routing protocol (e.g., RIP, IGRP)?

Answer Choices:

A. It calculates the path cost based on latency and bandwidth.

B. It maintains a complete topology map of the entire network.

C. It relies on neighbors' advertised routing tables and uses distance metrics (e.g., hop count) to choose the best path.

D. It converges instantly upon a topology change.

Question 16 (OSPF)

In an OSPF network, what is the special identifier given to the central, backbone area that connects all other areas?

Answer Choices:

A. Area ID 1

B. Area Border Router

C. Area ID 0 (Backbone Area)

D. Stub Area

Question 17 (Layer 2 Switching - STP)

What is the default priority number used by switches running the Spanning Tree Protocol (STP) before any manual configuration?

Answer Choices:

A. 1

B. 32768

C. 16384

D. 4096

Question 18 (Layer 2 Switching - STP)

In an STP topology, a switch port that provides the lowest cost path from the segment to the root bridge is called a:

Answer Choices:

A. Root Port

B. Designated Port

C. Blocking Port

D. Root Bridge

Question 19 (VLANs)

What is the industry standard IEEE protocol used to tag Ethernet frames with VLAN information when they traverse a trunk link?

Answer Choices:

A. IEEE 802.3

B. IEEE 802.1Q

C. IEEE 802.1D

D. Cisco ISL

Question 20 (VLANs)

In Cisco's **VLAN Trunking Protocol (VTP)**, which mode allows a switch to update its own VLAN database and send those changes to other switches in the domain?

Answer Choices:

A. Client mode

B. Transparent mode

C. Server mode

D. Access mode

Question 21 (Inter-VLAN Routing)

What network device is **required** to allow communication between hosts connected to different, segmented VLANs?

Answer Choices:

A. An STP Root Bridge

B. A Layer 2 Switch

C. A Router or Layer 3 Switch

D. A VTP Client

Question 22 (NAT)

What type of Network Address Translation (NAT) is typically used to allow multiple hosts inside a private network to share a single public IP address for Internet access?

Answer Choices:

A. Static NAT

B. NAT overloading (PAT)

C. NAT $1:1$

D. Dynamic NAT (without overload)

Question 23 (IPv6)

An IPv6 address typically consists of a 64-bit network prefix and a 64-bit Interface ID (host part). How long is the entire IPv6 address in bits?

Answer Choices:

A. 32 bits

B. 64 bits

C. 128 bits

D. 256 bits

Question 24 (Router Management)

If a Cisco router has a persistent password issue, which component must be properly configured (often to $0\text{x}2102$) to ensure the device loads the configuration from NVRAM during boot?

Answer Choices:

A. Flash Memory

B. Configuration Register

C. Boot Loader

D. ROM Monitor

Question 25 (EIGRP)

Which specific metric component does Cisco's Enhanced Interior Gateway Routing Protocol (EIGRP) use that is not typically used by RIP or OSPF?

Answer Choices:

A. Hop count

B. Bandwidth and Delay

C. Priority

D. Administrative Distance

Question 26 (High-Speed Interconnects)

In **InfiniBand Architecture**, what is the term for a logical connection between a sender and a receiver, defined by a pair of Queue Pairs (QPs)?

Answer Choices:

A. Host Channel Adapter (HCA)

B. Switch

C. Subnet Manager

D. Queue Pair

Question 27 (RAID)

Which RAID level provides striping with parity, allowing one disk failure, and is commonly used for cost-effective performance and redundancy in non-critical applications?

Answer Choices:

A. RAID 0

B. RAID 1

C. RAID 5

D. RAID 10

Question 28 (Network File System)

Which file system allows clients to transparently access and manipulate files on a shared network resource as if they were local?

Answer Choices:

A. GPFS

B. Lustre

C. NFS (Network File System)

D. Ext3

Question 29 (Storage Architecture)

What differentiates **NAS (Network Attached Storage)** from **SAN (Storage Area Network)**?

Answer Choices:

A. NAS uses Fibre Channel; SAN uses Ethernet.

B. NAS provides file-level access; SAN provides block-level access.

C. NAS is less expensive; SAN is always local (DAS).

D. SAN can only be used by Windows hosts.

Question 30 (Parallel File System)

In the Lustre File System, which component is responsible for storing the actual file data, as opposed to the file metadata?

Answer Choices:

A. Metadata Target (MDT)

B. Object Storage Target (OST)

C. Lustre Client

D. Object Storage Server (OSS)

Difficult-Level Multiple-Choice Questions (10 Questions)

Question 31 (VLSM Calculation)

Given the network $192.168.1.0/24$. If you need a subnet (Subnet 2) that supports 52 hosts, what is the **first usable host address** in that subnet, assuming the subnets are assigned efficiently and sequentially?

Answer Choices:

A. $192.168.1.64$

B. $192.168.1.65$

C. $192.168.1.63$

D. $192.168.1.1$

Question 32 (Routing Protocols - EIGRP)

In EIGRP, what is the term for a route that is known to be down, but for which a successor has not yet been found, triggering a query process to neighbors?

Answer Choices:

A. Passive

B. Stuck-in-Active (SIA)

C. Feasible Successor

D. Successor

Question 33 (STP)

If two switches connected via a trunk link have the same priority number, which factor is used as the tie-breaker to elect the **Root Bridge**?

Answer Choices:

A. The switch with the lowest speed port.

B. The switch with the highest MAC address.

C. The switch with the lowest MAC address.

D. The switch with the highest number of active ports.

Question 34 (VLANs - VTP)

If a switch is configured as a **VTP Transparent** mode switch, what is its behavior regarding VLAN database changes and VTP advertisements?

Answer Choices:

A. It receives and stores VTP advertisements but does not forward them.

B. It ignores VTP advertisements, does not store them, and does not forward them.

C. It receives VTP advertisements, does not process them, but forwards them transparently to other switches.

D. It modifies the VTP configuration revision number before forwarding the advertisement.

Question 35 (OmniPath/Infiniband)

The primary flow control mechanism used in high-speed interconnects like InfiniBand and OmniPath is based on:

Answer Choices:

A. CSMA/CD, similar to Ethernet.

B. Windowing (like TCP).

C. Link-level credits or tokens at the hardware layer.

D. Time-Division Multiplexing (TDM).

Question 36 (RAID)

Which RAID level provides the highest read/write performance for large files but offers **zero fault tolerance**?

Answer Choices:

A. RAID 1

B. RAID 5

C. RAID 6

D. RAID 0

Question 37 (Parallel File System)

What major characteristic of **GPFS (General Parallel File System)** enables it to be used in cluster environments?

Answer Choices:

A. Its proprietary reliance on Fibre Channel SAN only.

B. Its capability for concurrent access from multiple nodes to the same file.

C. Its use of the old FAT32 file structure.

D. Its limitation to only store metadata on local disks.

Question 38 (IPv6 Addressing)

Given the IPv6 prefix $2001:0\text{db}8:1234::/48$. If the subnet ID is 0001, what is the resulting network prefix for this subnet?

Answer Choices:

A. $2001:0\text{db}8:1234:0001::/64$

B. $2001:0\text{db}8:1234:0001::/48$

C. $2001:0\text{db}8:1234::1/64$

D. $2001:0\text{db}8:0001::/48$

Question 39 (Storage Devices)

What is the primary technical reason why **Solid State Drives (SSDs)** generally exhibit faster random access times compared to traditional **Hard Disk Drives (HDDs)**?

Answer Choices:

A. SSDs use a faster SATA interface.

B. SSDs have no mechanical moving parts (platters and read/write heads).

C. SSDs are always formatted with the XFS file system.

D. HDDs suffer from higher electromagnetic interference.

Question 40 (OSI Model)

Which layer of the OSI model is responsible for providing data compression and encryption/decryption services?

Answer Choices:

A. Network Layer

B. Presentation Layer

C. Session Layer

D. Application Layer
